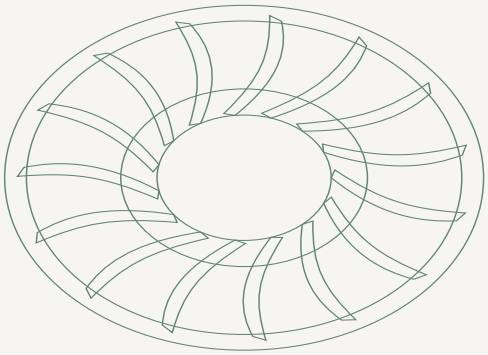
Harshavarthan Rajkumar.

MECHANICAL ENGINEERING / ROBOTICS / AI

From physical systems to intelligent machines.

Mechanical Engineering student at VIT bringing mechanical design, robotics and applied AI together through hands-on engineering projects.



KJ66 / CONCEPT SCHEMATIC

1st place

Medical Device AI Hackathon

Finalist

Honeywell Hackathon 2026

VIT / 2027

Expected B.Tech graduation

EXPLORE THE LIVE PORTFOLIO

Projects, technical details and source repositories.

https://harshavarthan6767.github.io

LinkedIn GitHub

Engineering the physical world.

01 KJ66 micro-turbojet diffuser

ONGOING DESIGN

Designed a 260 mm outer-diameter diffuser with five 72° LPBF segments, 15 vanes and a 6 mm floor. Developed a parametric build and a hybrid monolithic variant, with CFD as part of the ongoing workflow.

CadQuery / Fusion 360 / CFD / LPBF

Design in progress; specifications reflect the current concept.

02 Robotic arm systems

HARDWARE & SIMULATION

Built a D-H inverse-kinematics solver with 3D target visualization. Developed servo-driven arm work spanning gesture control, live ESP32-CAM telemetry and Arduino PWM control across 6-DOF / 7-DOF systems.

Python / Arduino / Unity / ESP32

Explore the repository

03 2D truss FEA solver

PUBLIC REPOSITORY

Implemented the direct stiffness method with coordinate transforms, pinned and roller supports, nodal displacements, reaction forces and axial stresses. Added deformed-shape plots, tension/compression colouring and CSV export.

Python / NumPy / FEA

Explore the repository

From sensing to decisions.

HONEYWELL HACKATHON / FINALIST / 2026

Three approaches to autonomous systems

GPS-denied navigation: 16-state EKF/VIO approach toward a 22-state EKF; target of <2 m drift over 12 km.
Drift interception: AirSim + YOLOv8 thermal targeting.
Swarm mapping: proposed 4-8 UAV mesh swarm with distributed SLAM and a 95% coverage target.

Drift and coverage figures are design targets, not measured field results.

Explore the swarm-mapping repository

CADForge CLI

IN DEVELOPMENT

Developing a vision-to-JSON specification pipeline, CadQuery generation and geometry-verification gates for STEP/STL output. The emphasis is on editable geometry and explicit checks throughout the generation process.

Autonomous drone control

PUBLIC REPOSITORY

The repository documents MAVLink communication, sensor fusion, wind estimation, PID position control and telemetry logging, with simulation examples for square missions and position hold. Includes geofencing and return-to-launch logic.

Explore repository

Cross-modal satellite retrieval

RESEARCH PIPELINE

A shared ViT-S/16 encoder, InfoNCE contrastive learning and a FAISS index form the retrieval pipeline. Supports top-5/top-10 search and ONNX export. Repository documentation identifies training and retrieval evaluation as pending.

Explore repository

A multidisciplinary foundation.

FIRST PLACE / AFFORDABLE MEDICAL DEVICE AI HACKATHON

Built an AI-assisted CTG interpretation tool to flag fetal distress. Won first place among multi-college teams in June-July 2024. Organized with AIIMS, TBI-VIT, MeitY, CMC Vellore and IEEE. Hackathon prototype.

Experience

Sundaram-Clayton Limited (TVS Group)

Industrial Process Insight Intern / May-Jun 2025

Observed cross-functional manufacturing operations and mechanical-finance coordination. Worked with paperless transactions, digital documentation and process automation.

Video editing & visual communication

IEEE Computer Society - VIT / Aug 2025-present
Web Wizard / Aug 2024-Oct 2025

Education & technical toolkit

VIT, Vellore: B.Tech Mechanical Engineering Core, 2023-2027.
Expected graduation: 2027. CGPA: 7.72/10.

Design: Fusion 360, SolidWorks, CadQuery, FEA, CFD, MATLAB.
Robotics: ESP32, Arduino, ROS2 simulation, EKF, kinematics, SLAM concepts.
Software & AI: Python, Java, FastAPI, Google ADK, CrewAI, NumPy, MongoDB, CI/CD, prompt engineering and AI integration.

Beyond engineering: State-level kickboxing bronze medal, photography and visual editing.

LET'S BUILD SOMETHING USEFUL.

LinkedIn / connect GitHub / explore

https://harshavarthan6767.github.io